PAIR-SV Train15 interim report

#Abstract

This interim validation experiment evaluates PAIR-SV on a frozen multisample snapshot of 15 training individuals and five validation individuals (29,924 and 9,981 successful sample–candidate examples). It is **not** the final 40/5/5 experiment and no internal-test person was accessed. Controlled fixed-seed M0, M2 and M4 runs use identical examples, five epochs, train-only normalization and validation-only threshold selection. M4 has the highest F1, but its margin over M2 is small; conclusions therefore remain cautious.

#Motivation, data and split

The question is whether paired breakpoint images benefit from sample-specific read support and candidate-aware REF/ALT representation. Positives are PanGenie GT 0/1 or 1/1 high-quality sequence-resolved deletions; negatives are same-site callable 0/0 genotypes. This design holds the candidate locus fixed across people and tests genotype-specific evidence. The person split is frozen: 15 train and five validation individuals here; internal test remains untouched.

#Extraction, QC and protocol

Each example has two `(6,64,128)` pileups, selected read sequences, 17 scalar support features and deduplicated GRCh38 REF/ALT context. Validation extraction passed 9,981/10,000 rows (0.19% failure); train shards used for this snapshot passed their QC. No sample ID, population, GQ, AF, negative type, source filename or split is passed to the model. M0 uses images, M2 adds scalars, and M4 adds REF/ALT and read embeddings. All three models were retrained with seed 20260720, five epochs, identical rows and a threshold maximizing validation F1.

#Main results

model	validation_loss	threshold	precision	recall	f1	roc_auc	pr_auc	mcc	tn	fp	fn	tp	epochs	training_seconds	parameters
M0	0.4261	0.415	0.7609	0.8413	0.7991	0.8836	0.8947	0.5807	3678	1318	791	4194	5	1065.28	287618
M2	0.4242	0.445	0.7591	0.8411	0.798	0.8844	0.8954	0.578	3665	1331	792	4193	5	1025.04	323110
M4	0.4102	0.435	0.7668	0.8497	0.8062	0.8914	0.9003	0.5954	3708	1288	749	4236	5	1044.46	622374

The controlled M4 F1 advantage over M2 is small. Bootstrap intervals below should not be read as final-test inference; rows are paired observations and candidate reuse limits independence.

#Paired bootstrap (1,000 row resamples)

comparison	n_bootstrap	f1_difference	f1_ci_low	f1_ci_high	f1_win_proportion	pr_auc_difference
M2 − M0	1000	-0.001	-0.005	0.0034	0.35	0.0008
M4 − M0	1000	0.007	0.0016	0.0124	0.995	0.0056

| M4 − M2 | 1000 | 0.0081 | 0.0029 | 0.0138 | 1 | 0.0049 |

#Branch dependence

intervention	validation_loss	threshold	precision	recall	f1	roc_auc	pr_auc	mcc	tn	fp	fn	tp	f1_change	roc_auc_change	pr_auc_change	mean_abs_logit_change	changed_binary_predictions
candidate_zero	0.419	0.435	0.7375	0.8736	0.7998	0.8894	0.8988	0.5731	3446	1550	630	4355	-0.0063	-0.002	-0.0015	0.164	0.0486
read_zero	0.425	0.435	0.7509	0.8514	0.798	0.8832	0.8938	0.5746	3588	1408	741	4244	-0.0082	-0.0082	-0.0065	0.1745	0.0549
scalar_zero	0.4416	0.435	0.7314	0.86	0.7905	0.8756	0.8842	0.5534	3422	1574	698	4287	-0.0156	-0.0158	-0.0161	0.3291	0.0815
candidate_shuffle	0.4221	0.435	0.767	0.8367	0.8003	0.885	0.8946	0.5855	3729	1267	814	4171	-0.0058	-0.0064	-0.0056	0.2046	0.0555
read_shuffle	0.4057	0.435	0.7716	0.8568	0.812	0.8952	0.9038	0.6074	3732	1264	714	4271	0.0058	0.0039	0.0035	0.2362	0.077
scalar_shuffle	0.3639	0.435	0.7833	0.9091	0.8415	0.9247	0.9291	0.6667	3742	1254	453	4532	0.0354	0.0333	0.0288	0.3634	0.115

Zero ablations remove an encoded branch; shuffle ablations break row correspondence. Non-zero logit and prediction changes demonstrate that M4 uses the branches. A shuffled read branch can occasionally improve a finite validation realization; that is not evidence that reads are harmful.

#Subgroups

Subgroup counts and metrics are in `pairsv_train15_subgroup_metrics.csv`. Small GT 1/1 or per-person strata require caution. The same-site-negatives row is an all-negative slice, so binary precision/recall/F1 are undefined there (reported as NaN); `specificity` (TN/(TN+FP)) is the informative metric for that row: M0 0.736, M2 0.734, M4 0.742. The short-deletion comparison is shown in the figure.

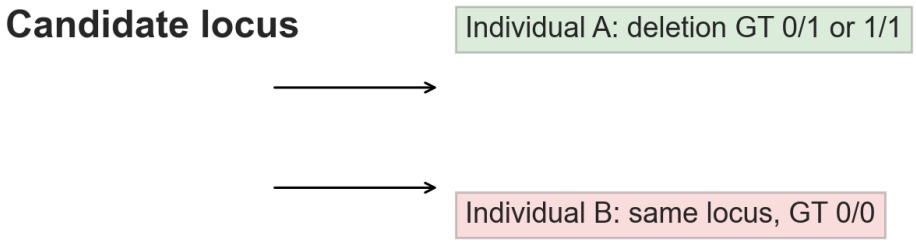
#Limitations and interim conclusion

Multisample dual-breakpoint models reach validation F1 around 0.80. Scalar read support is useful relative to controlled M0. Full PAIR-SV uses its additional branches and may improve short-deletion recall, but it has not yet shown a convincing overall advantage over M2. Remaining work: audit/recover all remaining shards, finish 40-person training, run three seeds, freeze threshold policy, evaluate once on the untouched five-person internal test, and perform cluster-aware inference. The final success target remains mean internal-test F1 ≥0.72 with consistent three-seed direction and multimodal dependence.

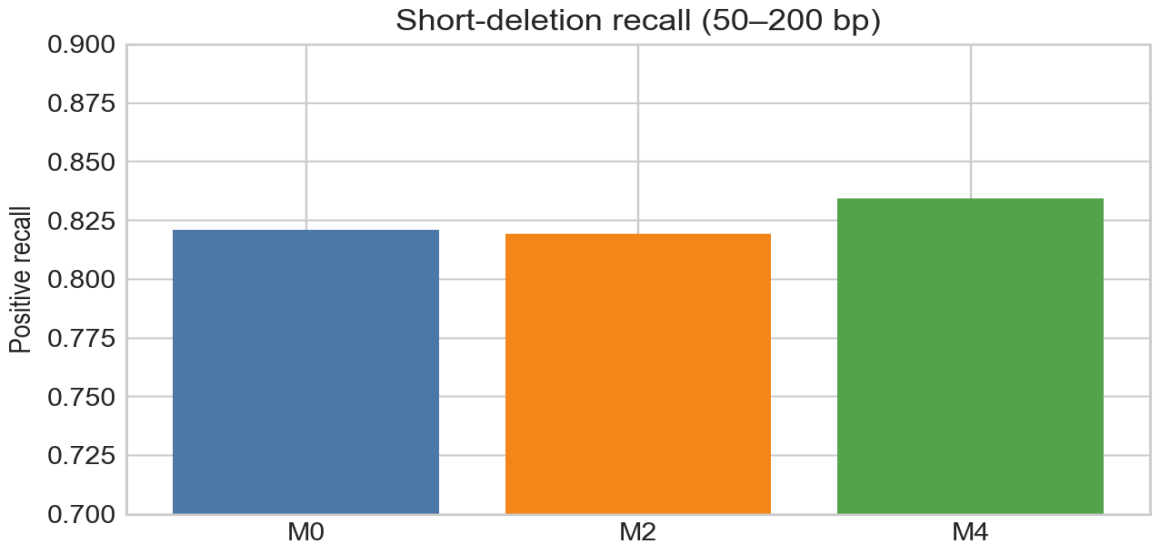
PAIR-SV architecture

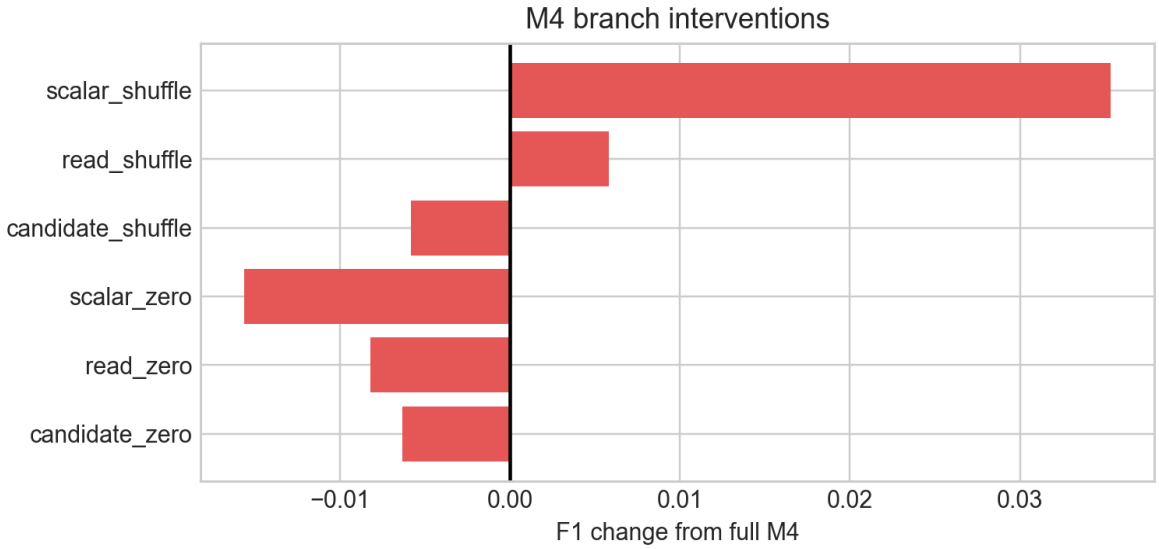


Same-site genotype-level design




Controlled Train15 comparison
Validation F1
0.83
0.82
0.81
0.80
0.79
0.78
0.77
0.76
0.75
0.799
0.798
0.806
M0
M2
M4


Short-deletion recall (50–200 bp)
Positive recall
0.900
0.875
0.850
0.825
0.800
0.775
0.750
0.725
0.700
M0
M2
M4


M4 branch interventions
scalar_shuffle
read_shuffle
candidate_shuffle
scalar_zero
read_zero
candidate_zero
−0.01
0.00
0.01
0.02
0.03
F1 change from full M4